

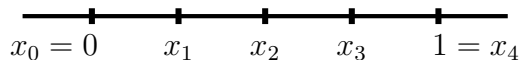
MAT 242. Riemann Sums

We will investigate the area under $f(x) = x^3$ on the interval from $x = a$ to $x = b$.

1. Begin with right endpoint Riemann Sums for area under $f(x) = x^3$ from $x = 0$ to $x = 1$.

- Divide $[0, 1]$ into 4 segments as indicated.

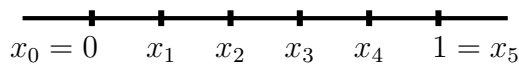
What are x_1, x_2, x_3 , and Δx ?



Write the right endpoint Riemann sum for $f(x) = x^3$ using this division of $[0, 1]$.

- Divide $[0, 1]$ into 5 segments as indicated.

What are x_1, x_2, x_3, x_4 , and Δx ?

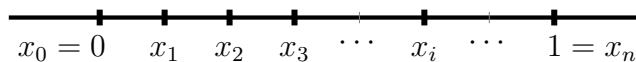


Write the right endpoint Riemann sum for $f(x) = x^3$ using this division of $[0, 1]$.

- Now divide $[0, 1]$ into n segments.

What are x_1, x_2, x_3 , and Δx ?

What is the i th value x_i ?



Use summation notation to write the right endpoint Riemann sum for $f(x) = x^3$.

Use the formula $\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2$ to express this without a summation.

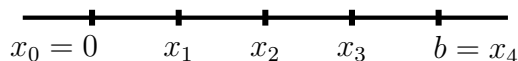
Area is $\lim_{n \rightarrow \infty}$ of the expression above.

Use your knowledge from last semester to compute this limit!

2. Now look at right endpoint Riemann Sums for area under $f(x) = x^3$ from $x = 0$ to $x = b$.

- Divide $[0, b]$ into 4 segments as indicated.

What are x_1, x_2, x_3 , and Δx ?

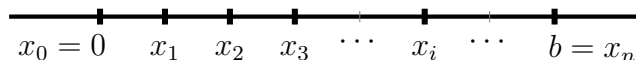


Write the right endpoint Riemann sum for $f(x) = x^3$ using this division of $[0, b]$.

- Now divide $[0, b]$ into n segments.

What are x_1, x_2, x_3 , and Δx ?

What is the i th value x_i ?



Use summation notation to write the right endpoint Riemann sum for $f(x) = x^3$.

Use the formula $\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2$ to express this without a summation.

Area is $\lim_{n \rightarrow \infty}$ of the expression above.

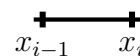
Use your knowledge from last semester to compute this limit!

Summary. Dividing interval $[a, b]$ into n segments yields:

$$\Delta x = \frac{b-a}{n}$$

$$\begin{aligned} x_0 &= a \\ x_1 &= a + \Delta x \\ x_2 &= a + 2 \Delta x \\ &\vdots \\ x_i &= a + i \Delta x \\ &\vdots \\ x_n &= a + n \Delta x = b \end{aligned}$$

Segment i of the division is:



Right endpoint Riemann sum:

$$\sum_{i=1}^n f(x_i) \Delta x$$

Left endpoint Riemann sum:

$$\sum_{i=1}^n f(x_{i-1}) \Delta x$$